

Exercice : Robustness of MIMO systems

Let us consider the system defined by the following block diagram.

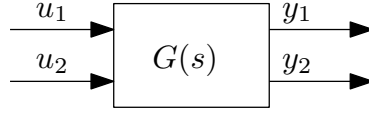


FIGURE 1 – Block Diagram of the studied system

As an example of a MIMO system, we can think about two motors sharing a common load, where the control inputs are the inputs of each motor, and the outputs are the load's velocity and the shaft tension (torque). Another example is an air-conditioning system, where two outputs are the air's temperature and velocity, and the control inputs are the inputs of a fan and of a heater. In both examples, the subsystems are interconnected, and the overall model is a MIMO 2-inputs-2-outputs system.

When linearized around a specific operation point, the model can be described with a 2×2 transfer matrix,

$$\begin{bmatrix} Y_1(s) \\ Y_2(s) \end{bmatrix} = \begin{bmatrix} W_{11}(s) & W_{12}(s) \\ W_{21}(s) & W_{22}(s) \end{bmatrix} \begin{bmatrix} U_1(s) \\ U_2(s) \end{bmatrix},$$

or a state-space representation

$$\begin{aligned} \dot{x} &= Ax + B \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}, \\ \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} &= Cx. \end{aligned}$$

Part 1 - Feedback Design

Suppose that

$$\begin{aligned} W_{11}(s) &= \frac{2}{(0.5s + 1)(0.7s + 1)}, & W_{12}(s) &= 0, \\ W_{21}(s) &= \frac{-0.3}{1.5s + 1}, & W_{22}(s) &= \frac{0.7}{(2s + 1)(0.5s + 1)}. \end{aligned}$$

Some useful Matlab commands : `tf`, `ss`, `place`, `lqr`.

1. Give the state-space representation of the system.
2. Design a state observer and an LQ feedback with the integral action,

$$u = -K_x \hat{x} + K_i \int_0^t (y^r(\tau) - y(\tau)) d\tau,$$

where $y^r = [y_1^r \ y_2^r]^\top$. Tune Q and R so that the transit time is of the order of a few seconds for both outputs. Choose observer's poles according to the LQ design.

3. Perform simulations to validate your design.
4. Rewrite the controller in the form $U(s) = R(s)Y^r(s) - K(s)Y(s)$, where $R(s)$ and $K(s)$ are transfer matrices; see Fig. 2a, where u , y , and y^r are vectors with two components. Verify in simulation the equivalence of this transformation.

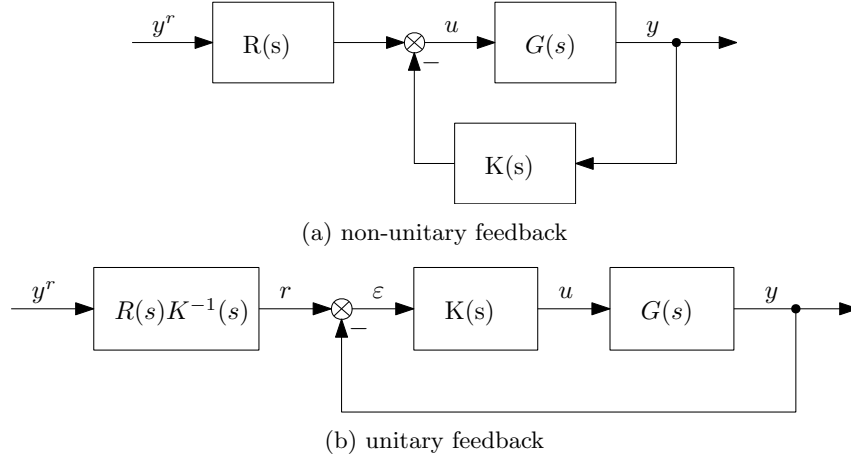


FIGURE 2 – Control structure

Part 2 - Frequency Domain Analysis

For the frequency-domain analysis it is often convenient to rewrite the system in the unitary feedback form; see Fig. 2b. Obviously, the closed-loop stability and robustness properties depend only on the transfer matrices $G(s)$ and $K(s)$; the transfer matrix $R(s)$ affects the transient performance but not the closed-loop stability. For simplicity, in what follows we will consider the structure given in Fig. 2b and the signal r as a new input.

Some useful Matlab commands : `feedback`, `sigma`, `freqresp`, `svd`, `norm(system,inf)`.

For the following manipulations, consider the frequency range from 10^{-2} till 10^2 rad/s.

1. Plot the singular values of the open-loop transfer matrix $G(s)K(s)$.
2. What are the high-/low-gain frequency zones? How do they correspond to the regulation performance?
3. Find the sensitivity transfer function $S(s)$ from the input r to the tracking errors ε and the complementary sensitivity function $T(s)$ from the input r to the output y .
4. Based on the open-loop analysis, make an approximate conclusion about the behavior of $S(s)$ and $T(s)$ for low and hi frequencies.
5. Plot the singular values of S and T . Are they consistent with the previous conclusions?
6. Find $\|T(s)\|_\infty$. To which frequency does it correspond? Confirm your results performing the SVD decomposition of $T(i\omega)$ for the chosen frequency, where $i = \sqrt{-1}$.

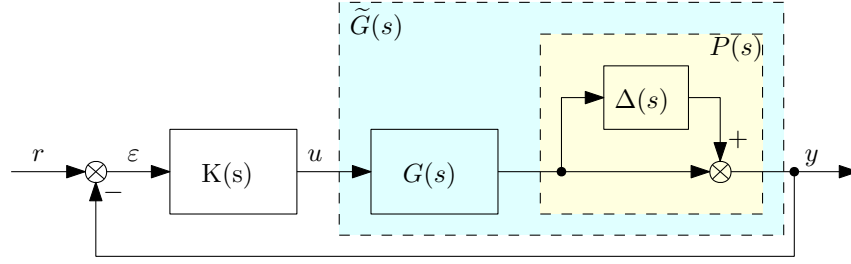
Part 3 - Robustness analysis

The actual transfer function $\tilde{G}(s)$ is uncertain and may differ from the model $G(s)$. One common way to represent his uncertainty is

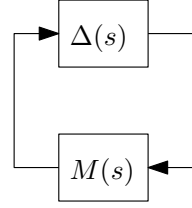
$$\tilde{G}(s) = G(s) \underbrace{(I + \Delta(s))}_{P(s)},$$

where $\Delta(s)$ is an unknown transfer matrix; see Fig. 3a.

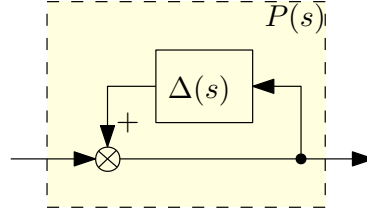
1. Find the transfer matrix $M(s)$ corresponding to Fig. 3b.
2. Applying the small-gain theorem, what should be a restriction for $\Delta(s)$ to guarantee the closed-loop stability?
3. Consider the gain uncertainty $P(s) = \begin{bmatrix} a_1 & 0 \\ 0 & a_2 \end{bmatrix}$. What are the admissible ranges for a_1 and a_2 ?



(a) Uncertainty $\Delta(s)$ in $\tilde{G}(s)$



(b) Equivalent transformation with $r \equiv 0$



(c) Another representation of the uncertainty

FIGURE 3 – Uncertainty in the model

4. Consider the phase uncertainty $P(s) = \begin{bmatrix} e^{j\phi_1} & 0 \\ 0 & e^{j\phi_2} \end{bmatrix}$. What are the admissible ranges for ϕ_1 and ϕ_2 ?
5. Suppose now that $P(s)$ has the form as in Fig. 3c. What is $M(s)$ in the equivalent transformation?
6. Consider the same gain and phase uncertainties for $P(s)$ as before. Are the admissible intervals the same?